1. SIGNAL GENERATION AND BASIC SIGNAL OPERATIONS

Objective:

This experiment is divided into four simulations:

1.Generation of Continuous Time Signals

2. Generation of Discrete Time Signals

3. Generation of Triangular and Sawtooth Signals

4. Step and Ramp Functions (Continuous & Discrete)

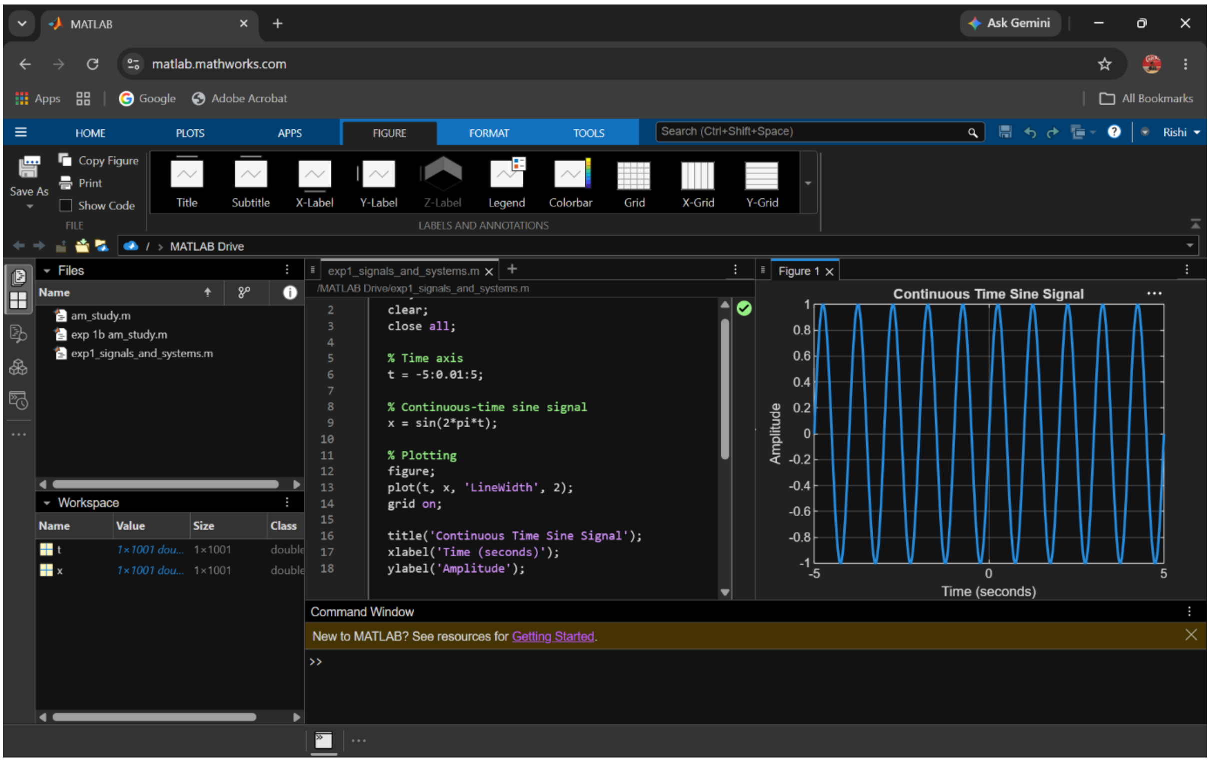
Experiment 1.1:

Generation of Continuous Time Signals

```
% Generation of Continuous Time Signals
clc;
clear;
close all;
% Time axis
t = -5:0.01:5;
% Continuous-time sine signal
x = sin(2*pi*t);
% Plotting
figure;
plot(t, x, 'LineWidth', 2);
grid on;

title('Continuous Time Sine Signal');
xlabel('Time (seconds)');
ylabel('Amplitude');
```

Output:



Exp 1.2:

% Generation of Discrete Time Signal

clc;

clear;

close all;

% Discrete-time index

n = -10:10;

% Discrete-time sinusoidal signal

x = sin(0.2*pi*n);

% Plotting

figure;

stem(n, x, 'filled');

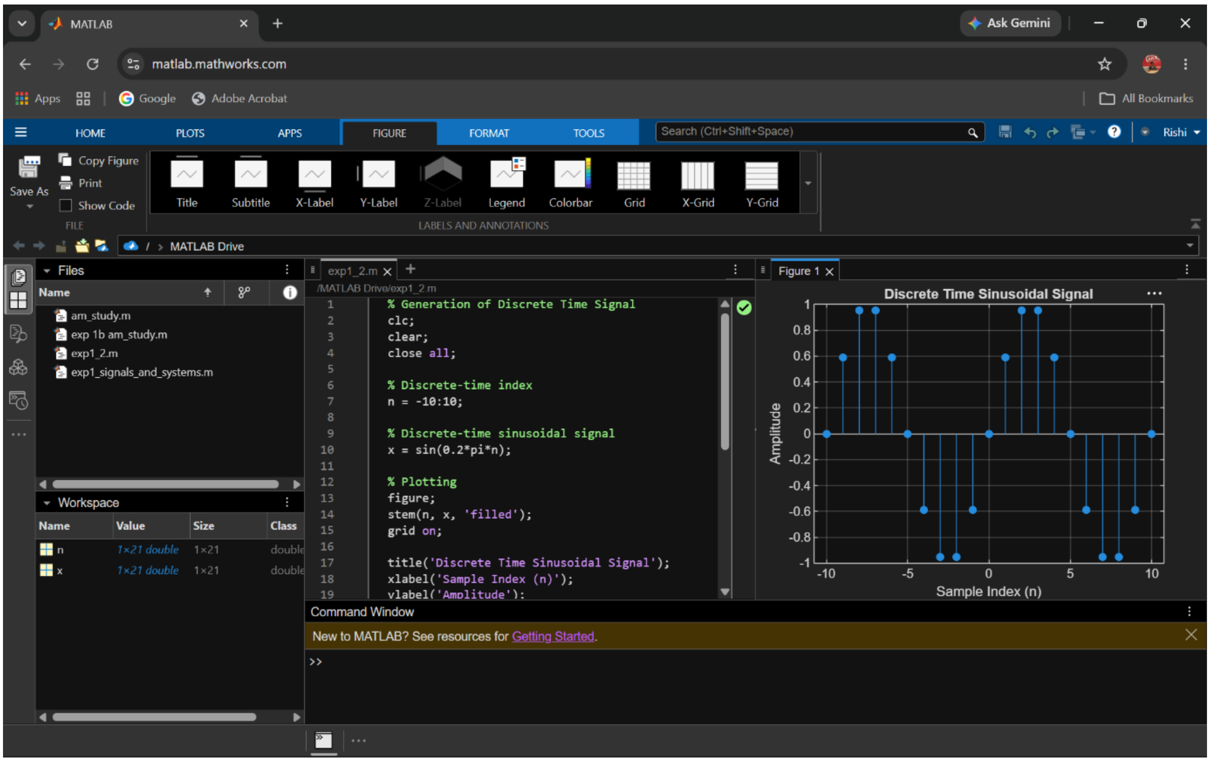
grid on;

title('Discrete Time Sinusoidal Signal');

xlabel('Sample Index (n)');

ylabel('Amplitude');

Output:



2. SIGNAL PROPERTIES AND TRANSFORMATIONS

Objective:

This experiment is divided into four simulations:

1. Periodic and Aperiodic Signals

2. Verification of Even and Odd Signal Decomposition

3. Amplitude and Time Scaling of Signals

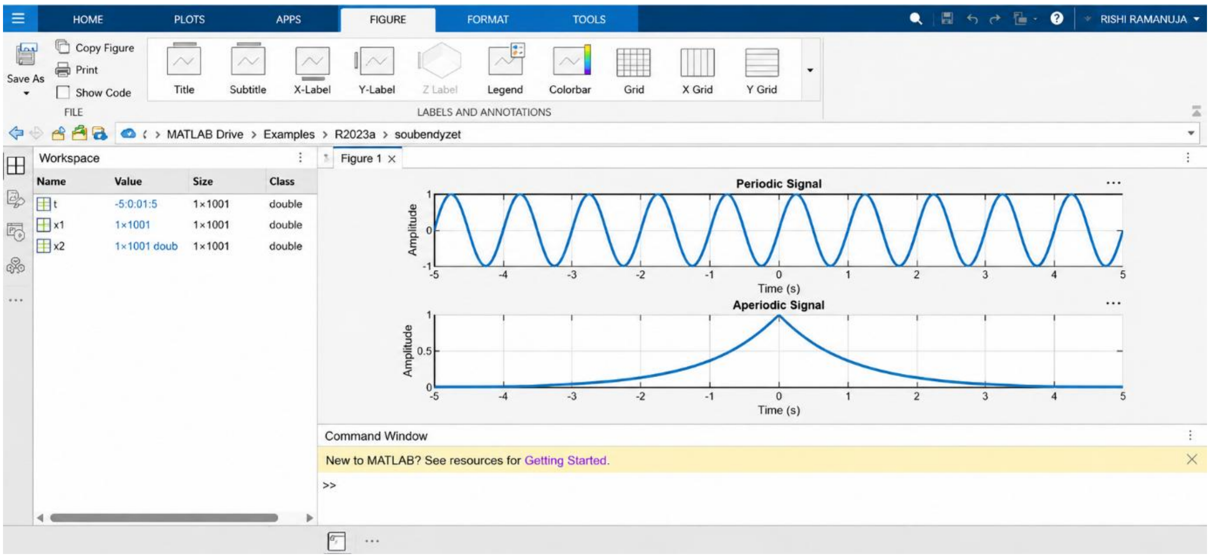
4. Time Shifting and Time Reversal of Signals

Experiment 2.1: Periodic and Aperiodic Signals

```
% Periodic and Aperiodic Signals
clc;
clear;
```

```
close all;
% Time axis
t = -5:0.01:5;
% Periodic Signal (Sine Wave)
x1 = sin(2*pi*t);
% Aperiodic Signal (Exponential Decay)
x2 = exp(-abs(t));
% Plotting
figure;
subplot(2,1,1);
plot(t,x1,'LineWidth',2);
grid on;
title('Periodic Signal');
xlabel('Time (s)');
ylabel('Amplitude');
subplot(2,1,2);
plot(t,x2,'LineWidth',2);
title('Aperiodic Signal');
xlabel('Time (s)');
ylabel('Amplitude');
```

Output:



Experiment 2.2: Verification of Even and Odd Signal Decomposition

```
% Verification of Even and Odd Signal Decomposition
clc;
clear;
close all;
% Time axis
t = -5:0.1:5;
% Original Signal
x = t.^3 + t;
% Time-reversed Signal
x_rev = (-t).^3 + (-t);
% Even Component
xe = (x + x_rev)/2;
% Odd Component
xo = (x - x_rev)/2;
% Reconstruction of Original Signal
x_reconstructed = xe + xo;
% Plotting
figure;
subplot(4,1,1);
plot(t,x,'LineWidth',2);
grid on;
title('Original Signal x(t)');
xlabel('Time');
ylabel('Amplitude');
subplot(4,1,2);
plot(t,xe,'LineWidth',2);
grid on;
title('Even Component x_e(t)');
```

```
xlabel('Time');
ylabel('Amplitude');
subplot(4,1,3);
plot(t,xo,'LineWidth',2);
grid on;
title('Odd Component x_o(t)');
xlabel('Time');

ylabel('Amplitude');
subplot(4,1,4);
plot(t,x_reconstructed,'LineWidth',2);
grid on;
title('Reconstructed Signal x_e(t) + x_o(t)');
xlabel('Time');
ylabel('Amplitude');
```

Output: